Preprint

SUPPLEMENTARY MATERIAL

The Stellated Octahedron, the Cube and the Hadamard Matrix

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Abstract
A cube simplex is a set of $D+1$ vertices of the cube $\{-1,+1\}^D$ that are pairwise equidistant. We show that the coordinate matrix M of a cube simplex, bordered by a row of ones, is a Hadamard matrix, and conversely; that this is the forced Gram matrix of the tetrad frame, $M^TM=(D+1)I-J$; and that the Hadamard rows encode closure and an isotropic second moment. A cube simplex therefore exists only if $D+1$ is 2 or divisible by 4. A cube simplex and its parity image are disjoint, and together they fill the cube only for $D=3$, where they are the two parity classes $x_1x_2x_3=\pm1$ and their compound is the stellated octahedron. This is the content of the equation $2(D+1)=2^D$. In a stated ordering, the 8×8 Walsh matrix of the cube is exactly $H_2\otimes\hat{H}$, with H_2 acting as spatial parity and $\hat{H}$ the bordered frame. Only for $K=2$ and $K=4$ do the rows of M exhaust the ways of splitting the K points into equal halves.

Setting

Let $Q_D=\{-1,+1\}^D$, $D\geq1$, and $K=D+1$. A *cube simplex* is a set of K distinct points of Q_D that are pairwise equidistant. Write them as the columns $v_0,\dots,v_D$ of a $D\times K$ matrix M with entries ±1, let $\mathbf{1}\in\mathbb{R}^K$ be the all-ones vector, and let

$$\hat{H}=\begin{pmatrix}\mathbf{1}^T\\M\end{pmatrix}, \tag{1}$$

a $K\times K$ matrix of signs. A Hadamard matrix of order n is an $n\times n$ matrix H of signs with $H^TH=nI$; for a square matrix this is equivalent to $HH^T=nI$.

The Frame Is a Hadamard Matrix

Theorem 1. *For a $D\times K$ sign matrix M the following are equivalent:*

(a) *the columns of M form a cube simplex;*

(b) $M^TM=KI-J$*;*

(c) $\hat{H}$ *is a Hadamard matrix;*

(d) $M\mathbf{1}=0$ *and* $MM^T=K\,I_D$*.*

Proof. (a)⇒(b): every column has squared norm D, and equal distances between columns of equal norm mean equal inner products, so $M^TM=\alpha I+\beta J$ with $\alpha+\beta=D$. Distinct columns give $\alpha=\frac{1}{2}|v_a-v_b|^2>0$, so the eigenvalue α, of multiplicity $K-1=D$ on $\mathbf{1}^\perp$, makes the rank at least D. The columns lie in $\mathbb{R}^D$, so the eigenvalue $\alpha+K\beta$ on $\mathbf{1}$ vanishes. Hence $\beta=-1$ and $\alpha=K$; this is the forced Gram matrix of the tetrad frame with $c^2=D$ [1]. (b)⇔(c): $\hat{H}^T\hat{H}=J+M^TM$. (c)⇔(d): $\hat{H}\hat{H}^T$ has blocks K, $(M\mathbf{1})^T$, $M\mathbf{1}$ and MM^T. (b)⇒(a): $|v_a-v_b|^2=2K$ for $a\neq b$. □

Every Hadamard matrix of order K becomes one with first row $\mathbf{1}^T$ after negating columns, and its remaining rows then form a matrix M whose columns are a cube simplex. Cube simplices and normalised Hadamard matrices are therefore the same objects. Condition (d) is the pair of conditions imposed on the tetrahedral directions in Letter 1 [2]: no net direction, and no preferred axis.

Lemma (Hadamard). *A Hadamard matrix of order $n>2$ has $4\mid n$.*

Proof. Normalise the first row to $\mathbf{1}^T$. Let $n_{\pm\pm}$ count the columns by their signs in rows 2 and 3. Orthogonality of rows 2 and 3 to row 1 and to each other gives $n_{++}+n_{+-}=n_{-+}+n_{--}$, $n_{++}+n_{-+}=n_{+-}+n_{--}$ and $n_{++}+n_{--}=n_{+-}+n_{-+}$, so all four counts are equal and $n=4\,n_{++}$. □

Corollary 1. *A cube simplex in Q_D exists only if $D=1$ or $4\mid D+1$. The first case beyond the segment is $D=3$.*

Only Three Dimensions Fill the Cube

Theorem 2. *Let V be a cube simplex in Q_D, $D\geq2$. Then V and $-V$ are disjoint cube simplices, and $V\cup(-V)=Q_D$ if and only if $D=3$. For $D=3$, V and $-V$ are the two parity classes $\{x_1x_2x_3=+1\}$ and $\{x_1x_2x_3=-1\}$, and their compound is Kepler's stella octangula [3, 4], the stellated octahedron.*

Proof. Negation preserves inner products, so $-V$ is a cube simplex. A shared point $v_a=-v_b$ would need $v_a\cdot v_b=-D$, but by Theorem 1 the inner products are D and -1, and $D\geq2$. So $|V\cup(-V)|=2(D+1)$, which equals $|Q_D|=2^D$

only for $D = 3$: the equation fails at $D = 2$, and for $D \geq 4$, $2^D > 2(D+1)$ by induction from $16 > 10$. For $D = 3$, $v_a \cdot v_b = -1$ means that v_a and v_b differ in exactly two coordinates, so they have the same product $x_1x_2x_3$; the four points of V lie in one parity class of four, and $-V$ in the other. □

The equation $2(D+1) = 2^D$ counts the same set twice: on the left, the vertices of a simplex and its parity image; on the right, the vertices of the cube.

The Walsh Factorisation

The Walsh functions on Q_3 are $\chi_S(x) = \prod_{i \in S} x_i$, $S \subseteq \{1, 2, 3\}$, and the Walsh matrix is $W_{S,x} = \chi_S(x)$.

Theorem 3. *Let V be the parity class $x_1x_2x_3 = +1$, with columns $v_0, \ldots, v_3$ of M. Order the points of Q_3 as $v_0, \ldots, v_3, -v_0, \ldots, -v_3$, and the Walsh functions as 1, x_2x_3, x_1x_3, x_1x_2 (even degree), then $x_1x_2x_3$, x_1, x_2, x_3 (odd degree). Then*

$$W = H_2 \otimes \hat{H} = \begin{pmatrix} \hat{H} & \hat{H} \\ \hat{H} & -\hat{H} \end{pmatrix}, \tag{2}$$

where H_2 has rows $(1, 1)$ and $(1, -1)$.

Proof. On V, $x_1x_2x_3 = 1$, so $x_2x_3 = x_1$, $x_1x_3 = x_2$ and $x_1x_2 = x_3$: both lists restrict to $1, x_1, x_2, x_3$, whose values on $v_0, \ldots, v_3$ are the rows of $\hat{H}$. Since $\chi_S(-x) = (-1)^{|S|}\chi_S(x)$, the even functions take the same values on $-V$ and the odd functions the opposite values. □

The factor H_2 is spatial parity, $x \to -x$, acting on Walsh functions by their degree; the factor $\hat{H}$ is the frame. On each tetrahedron, every Walsh function coincides with its complement up to a common sign, so the eight functions reduce to four.

Why Four Legs

Corollary 2. *For a cube simplex with K even, each of the $K - 1$ rows of M splits the K points into two halves, and the rows give $K - 1$ distinct splits. They are all such splits if and only if $K = 2$ or $K = 4$.*

Proof. By Theorem 1(d), each row has as many $+1$ as -1 entries. Two rows never give the same split: they are orthogonal, and a row and its negative are not. There are $\frac{1}{2}\binom{K}{K/2}$ splits, equal to $K - 1$ at $K = 2$ and $K = 4$; for $K \geq 6$, $\frac{1}{2}\binom{K}{K/2} \geq \frac{1}{2}\binom{K}{2} > K - 1$. □

At $K = 4$ the three rows of M are the three ways of splitting the four points, the four legs of the frame, into two pairs.

Summary

On the cube, the regular simplex of the tetrad frame and the normalised Hadamard matrix are one object: its columns satisfy the forced Gram matrix, and its rows satisfy closure and isotropy. Hadamard's condition limits such simplices to $D = 1$ and $D + 1 \equiv 0 \pmod 4$. A simplex and its parity image fill the cube only in three dimensions, where they are the two parity classes and form the stellated octahedron; this is what $2(D+1) = 2^D$ counts. The Walsh matrix of the three-cube factorises exactly as spatial parity times the frame.

References

[1] S. Nelson, *The Spatial Geometry of the Lorentz Tetrad* (2026).

[2] S. Nelson, *Supplementary Material: From the Lorentz Signature to the Tetrahedral Frame* (2026).

[3] J. Kepler, *Harmonices Mundi* (Linz, 1619).

[4] P. R. Cromwell, *Polyhedra* (Cambridge University Press, Cambridge, 1997).